

Assignment on Deploying a MERN Application through Terraform and Ansible

Objective:

Gain practical experience in deploying a MERN stack application on AWS using infrastructure automation with Terraform and configuration management with Ansible.

MERN Application to use: <https://github.com/UnpredictablePrashant/TravelMemory>

Tasks:

Part 1: Infrastructure Setup with Terraform

1. AWS Setup and Terraform Initialization:

- Configure AWS CLI and authenticate with your AWS account.
- Initialize a new Terraform project targeting AWS.

2. VPC and Network Configuration:

- Create an AWS VPC with two subnets: one public and one private.
- Set up an Internet Gateway and a NAT Gateway.
- Configure route tables for both subnets.

3. EC2 Instance Provisioning:

- Launch two EC2 instances: one in the public subnet (for the web server) and another in the private subnet (for the database).
- Ensure both instances are accessible via SSH (public instance only accessible from your IP).

4. Security Groups and IAM Roles:

- Create necessary security groups for web and database servers.
- Set up IAM roles for EC2 instances with required permissions.

5. Resource Output:

- Output the public IP of the web server EC2 instance.

Part 2: Configuration and Deployment with Ansible

1. Ansible Configuration:

- Configure Ansible to communicate with the AWS EC2 instances.

2. Web Server Setup:

- Write an Ansible playbook to install Node.js and NPM on the web server.
- Clone the MERN application repository and install dependencies.

3. Database Server Setup:

- Install and configure MongoDB on the database server using Ansible.
- Secure the MongoDB instance and create necessary users and databases.

4. Application Deployment:

- Configure environment variables and start the Node.js application.
- Ensure the React frontend communicates with the Express backend.

5. Security Hardening:

- Harden the security by configuring firewalls and security groups.
- Implement additional security measures as needed (e.g., SSH key pairs, disabling root login).

Deliverables:

- Terraform scripts for AWS infrastructure setup.
- Ansible playbooks for configuration and deployment of the MERN application.

- A detailed report documenting the implementation process, including how the application components interact with each other and the infrastructure.
- Screenshots or a video recording demonstrating the working MERN application.

Submission Instructions:

To submit your assignment, please follow these guidelines:

- Ensure that your assignment is fully completed.
- Push your code/assignment to a GitHub repository.
- Share the repository link by including it in a text, Word, or PDF file format.
- Submit the file/text containing the repository link via Vlearn.